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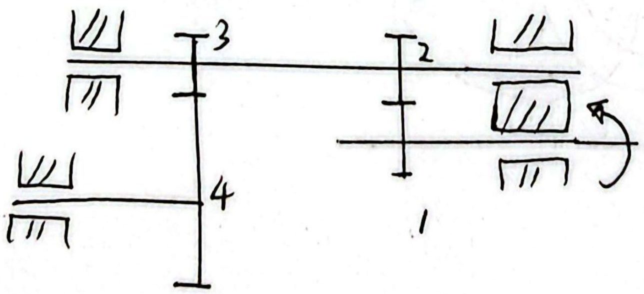
Q14-4 实现增力功能的机构

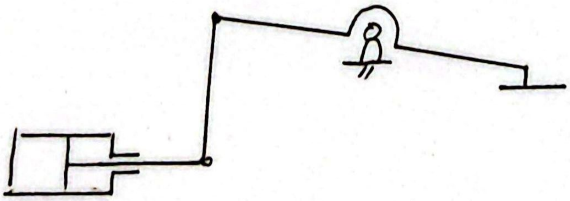
① （减速器）

小齿轮带大齿轮减速，转速降低扭矩（$\vec{T}=\vec{r}\times\vec{F}$）放大，实现增力

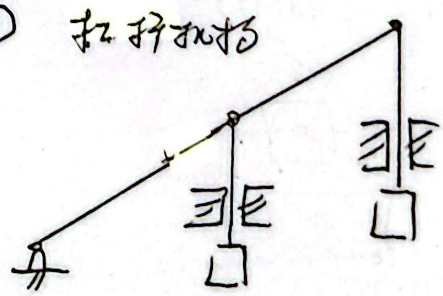


② 液压机构（千斤顶）

$$p=\frac{F_1}{A_1}=\frac{F_2}{A_2}$$



③ 杠杆机构



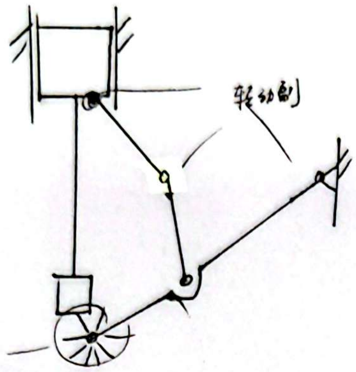
Q14.6 (冲压式蜂窝煤成型机) { 曲柄滑块 / 间歇运动

现添加清扫机构(冲头向上完成)

(已知) n = 60 rpm

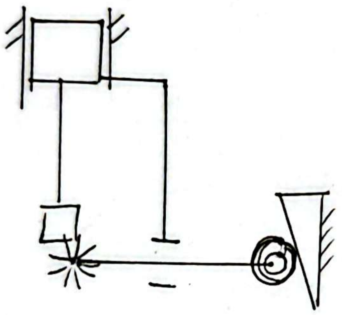
(1) 构思清扫机构方案(简图)

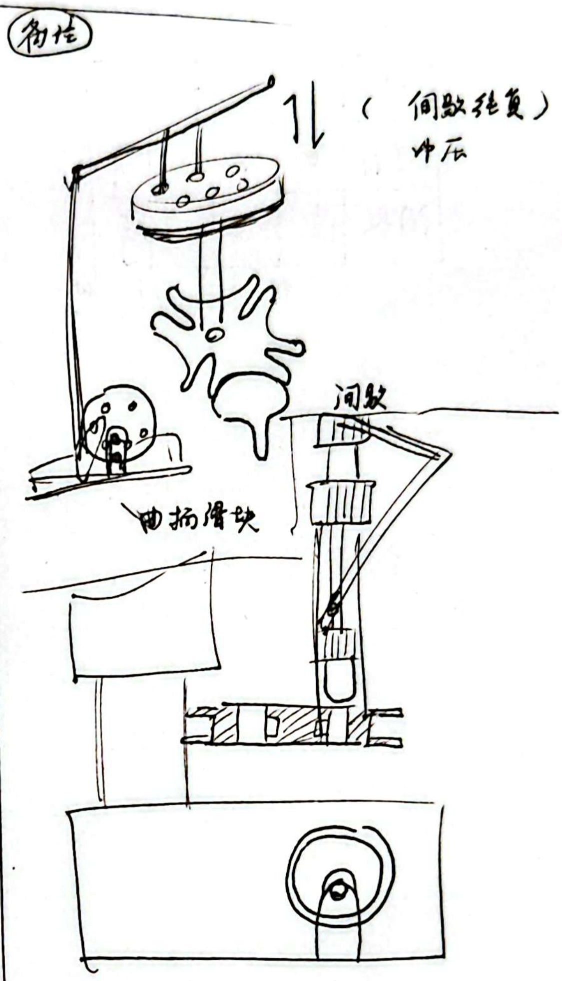
法①



扫刷机构

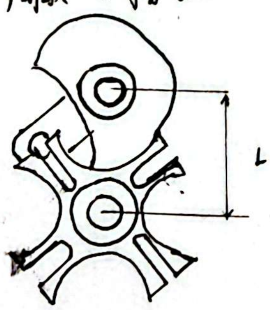
法②



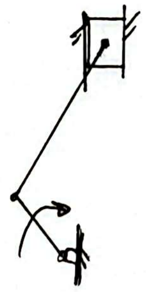


(2) 示意图(间歇,曲柄滑块,扫刷)

间歇(槽轮)



曲柄滑块



(3) (机械主轴)

曲柄轴(主轴) → 齿轮/链轮 → 间歇(送模筒)

↳ 清扫机构

(完成以下动作)

送料,冲压,清扫,复位

Q14-6

(3) 运动循环图

冲头	上		下	
清扫	停	动		停
间歇	停	动		停

0 90 180 270 360

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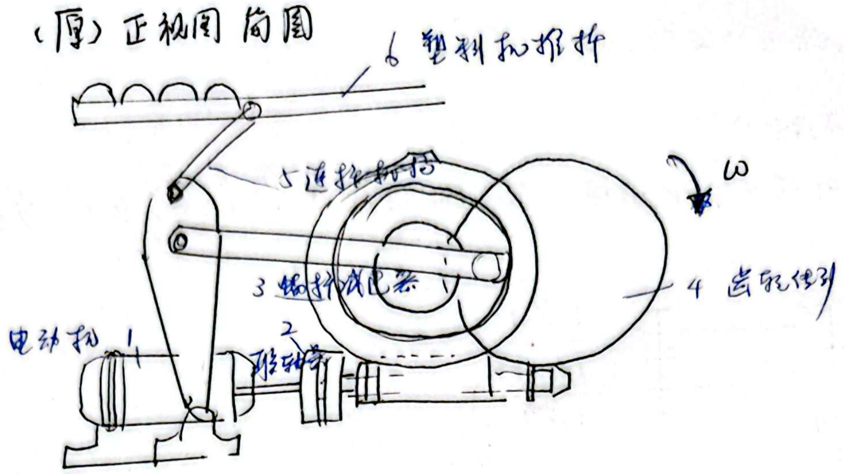
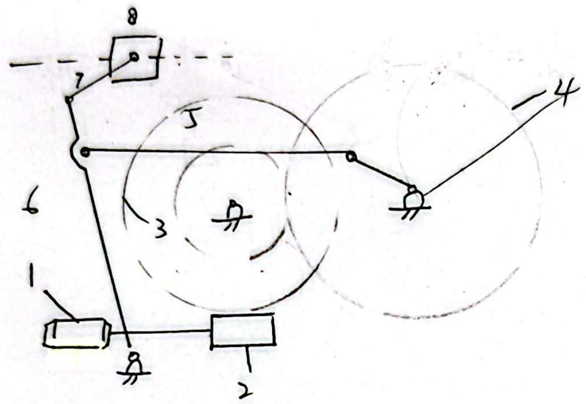
(原) 正视图 简图



(1) 简图



(2) 原动机：交流电动机，$P = 8kW$，~~$\omega = 1800\ rad/s$~~ $\omega = 1800\ rad/s$

(3) 蜗轮蜗杆 $i_{23} = 60$，$i_{34} = 5$

~~$i_{23} \times i_{34} = \frac{\omega_{motor}}{\omega}$~~

~~$\frac{\omega_{motor}}{\omega} =$~~

$i_{14} = \frac{n_1}{n_4} = \frac{34}{31}$

$\because i_{14} = \cancel{i_{12}} \cdot \cancel{i_{23}} \cdot \cancel{i_{34}} = 300$

满足 $\frac{n_{motor}}{n_4} = \frac{\omega_{motor}}{\omega_4} = \frac{\omega_{motor}}{\omega} = \frac{1800}{6} = 300$

(ω 随 ω_4 凸轮机构)

$\therefore$ 取 $\omega_{motor} = 1800\ rad/s$